APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: Geometry: Perimeter & Area of Rectangles

SUBJECT: Mathematics (MATH101)

CLASS & SECTION: Class 5 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: 89 / 100 (Grade A)

TEACHER DIRECTIVE & PROBLEM FOCUS:
Measure and calculate perimeter & area of 5 household objects with unit conversion steps.

1. Executive Overview & Problem Context

Two-dimensional Euclidean geometry distinguishes between linear boundary length (Perimeter, units: cm, m) and enclosed planar two-dimensional surface capacity (Area, units: cm², m²). This laboratory exercise applies empirical metric measurements across real-world rectangular household furnishings.

2. Core Analysis, Solutions & Systematic Working

- **Mathematical Formulations:**

  For a rectangle of length l and width w: Perimeter P = 2(l + w) = 2l + 2w. Area A = l × w. For a square with side s: P = 4s, A = s². Documented dimensional consistency throughout arithmetic steps.

- **Specimen 1: Student Study Desk:**

  Measured dimensions: Length l = 120 cm, Width w = 60 cm. Perimeter: P = 2(120 + 60) = 2(180) = 360 cm (3.60 meters). Area: A = 120 cm × 60 cm = 7,200 cm² = 0.72 m².

- **Specimen 2: Academic Mathematics Textbook:**

  Measured dimensions: Length l = 28 cm, Width w = 21 cm. Perimeter: P = 2(28 + 21) = 2(49) = 98 cm. Area: A = 28 × 21 = 588 cm².

- **Specimen 3: Bedroom Window Glass Pane:**

  Measured dimensions: Length l = 150 cm, Width w = 100 cm. Perimeter: P = 2(150 + 100) = 500 cm (5.0 m). Area: A = 150 × 100 = 15,000 cm² (1.50 m²).

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3. Detailed Data Matrix & Systematic Findings

Object Measured	Length (l)	Width (w)	Perimeter: 2(l+w)	Area: (l × w)
Study Table Surface	120 cm (1.2 m)	60 cm (0.6 m)	360 cm (3.6 m)	7,200 cm² (0.72 m²)
Mathematics Textbook	28 cm	21 cm	98 cm (0.98 m)	588 cm² (0.0588 m²)
Living Room Carpet	300 cm (3.0 m)	200 cm (2.0 m)	1,000 cm (10.0 m)	60,000 cm² (6.00 m²)
Bedroom Window Pane	150 cm (1.5 m)	100 cm (1.0 m)	500 cm (5.0 m)	15,000 cm² (1.50 m²)
Ceramic Floor Tile	40 cm (0.4 m)	40 cm (0.4 m)	160 cm (1.6 m)	1,600 cm² (0.16 m²)

4. Student Conclusions & Reflective Synthesis

Accurate differentiation between perimeter (fencing, framing, borders) and area (carpeting, painting, paving) is indispensable for practical civil engineering, architecture, and daily spatial budgeting.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Analytical Depth & Research Methodology: 28 / 30

Problem Solving & Structured Working: 18 / 20

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Very good effort. Clear and organized dimensional calculations. Unit conversions from cm to meters are sound."

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